
Digital Electronics (NECE 102)

(Latch and Flip Flop)

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combinational and sequential circuits

combinational circuits

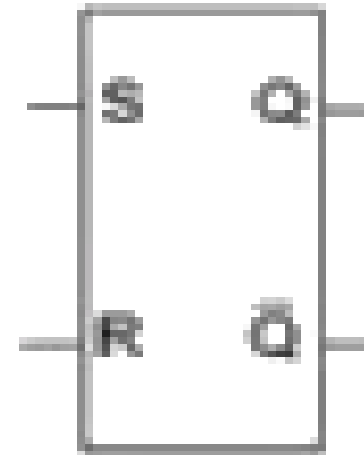
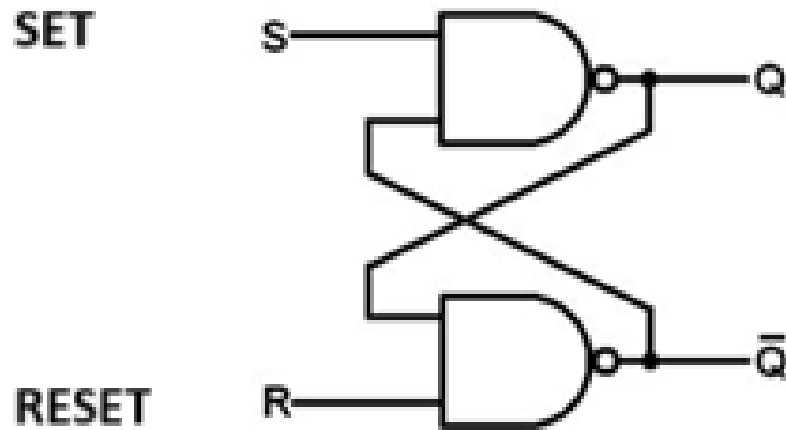
- Output depends only on the present inputs
- No memory involved
- No feedback
- Less complex/Easy to design
- Faster
- Eg: adder, subtractor, encoder, decoder, mux demux, code converter

sequential circuits

- Output depends on the present inputs and previous output
- Memory involved
- Feedback from output to input
- More complex because of feedback
- Slower
- Eg : Latches, flip-flop, register, counter etc

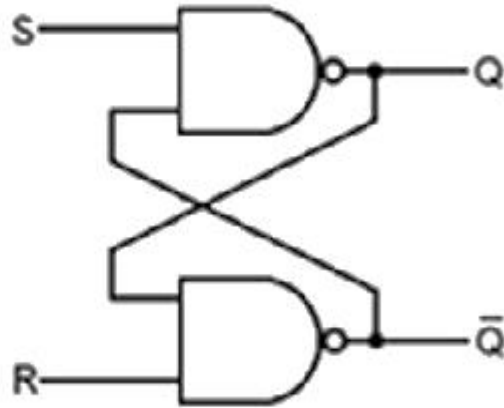
Latches

SR latch using NAND gate



SR Latches using NAND Gate

SET



RESET

- Case 1 – $S=0, R=0, Q=1$ & $Q'=1$
Invalid
- Case 2 – $S=1, R=0, Q=0$ & $Q'=1$
 $S=1, R=1, Q=0$ & $Q'=1$
- Case 3 – $S=0, R=1, Q=1$ & $Q'=0$
 $S=1, R=1, Q=1$ & $Q'=0$

NAND Truth table

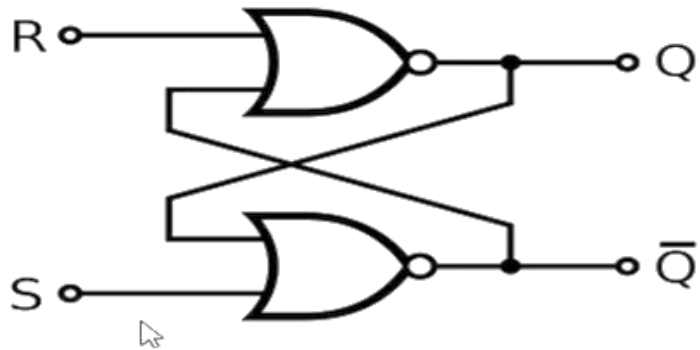
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

SR Latch Truth table

S	R	Q	Q'
0	0	Invalid	
0	1	1	0
1	0	0	1
1	1	No change	

$S = 0, R = 0$ then consider $S = 1$ and $R = 1$

SR Latches using NOR Gate



- Case 1 – $S=0, R=1, Q=0$ & $Q'=1$
 $S=0, R=0, Q=0$ & $Q'=1$
- Case 2 – $S=1, R=0, Q=1$ & $Q'=0$
 $S=0, R=0, Q=1$ & $Q'=0$
- Case 3 – $S=1, R=1, Q=0$ & $Q'=0$
 $S=0, R=0, Q=0$ & $Q'=1$
 $Q=1$ & $Q'=0$

NOR Truth table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

SR Latch Truth table

S	R	Q	Q'
0	0	No change	
0	1	0	1
1	0	1	0
1	1	Invalid	

$S = 1, R = 1$ then consider $S = 0$ and $R = 0$

Difference between Latch & flip-flop

Latch

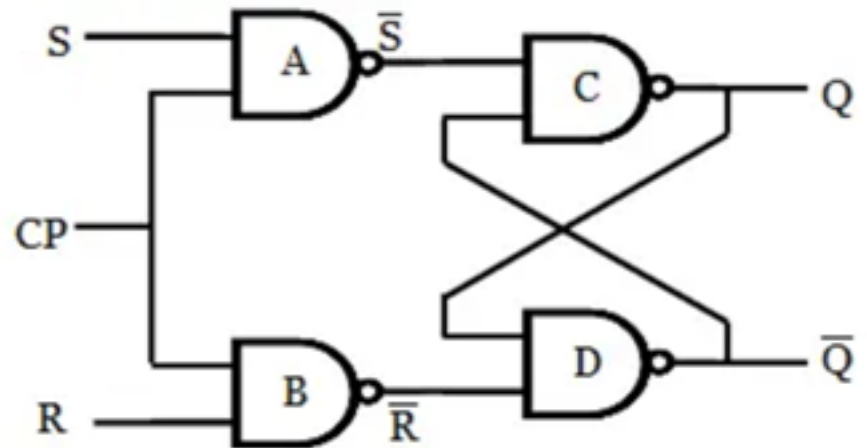
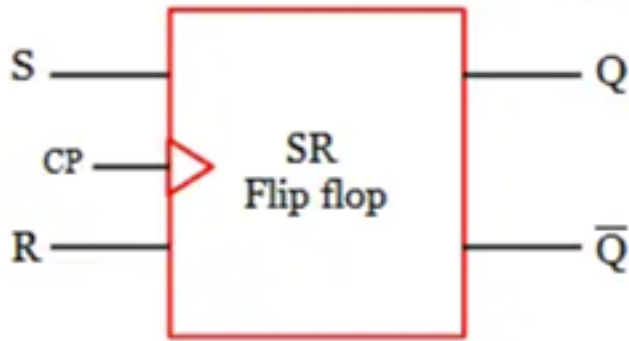
- It is based on logic gate
- It does not have clock in its internal circuit
- It changes output continuously based in the change of input
- Latch is sensitive to inputs
- Asynchronous circuit
- Required less power
- Less complex and faster

Flip-flop

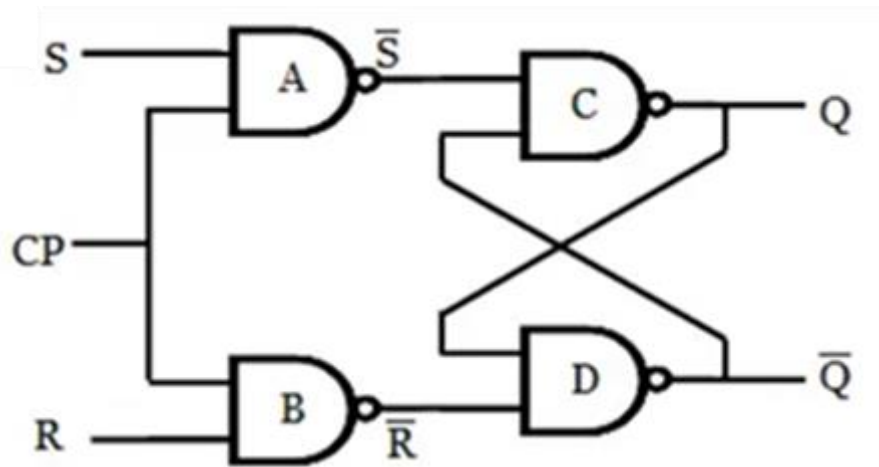
- It is based on latch and logic gates
- It has a clock in its internal circuit
- It changes output based on the change of input and clock
- Latch is sensitive to inputs and clock
- It is synchronous circuit
- Required more power
- More complex and slower

Flip-flop

Flip-Flop is an electronic component that is used to store one single bit of information.



SR Flip Flop using NAND gate



SR Latch Truth table

$\bar{S}$	$\bar{R}$	Q	Q'
0	0	Invalid	
0	1	1	0
1	0	0	1
1	1	No change	

CLK = 0; No change state

Truth Table of SR Flip Flop

CLK	S	R	Q	$\bar{Q}$
0	X	X	Hold	
1	0	0	Hold	
1	0	1	0	1
1	1	0	1	0
1	1	1	Invalid	

CLK	S	R	Q _{n+1}	State
0	X	X	Q _n	Hold
1	0	0	Q _n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	X	Invalid

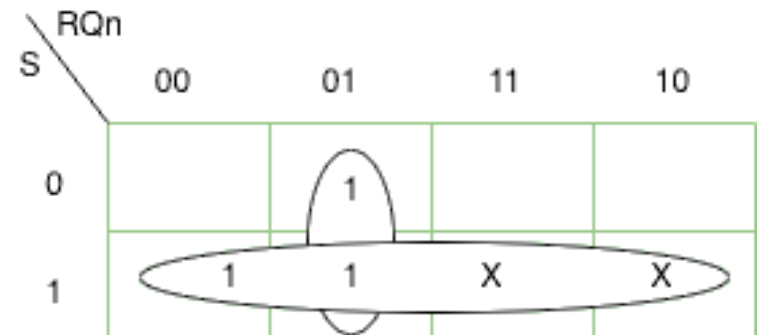
SR Flip Flop: Characteristic Equation

The characteristic equation tells us about what will be the next state of flip flop in terms of present state.

S	R	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Truth Table of SR Flip Flop

S	R	Q _{n+1}	State
0	0	Q _n	Hold
0	1	0	Reset
1	0	1	Set
1	1	X	Invalid



$$Q_{n+1} = S + R' Q_n$$

SR Flip Flop: Excitation table

Excitation Table tells about the excitation which is required by flip flop to go from current state to next state.

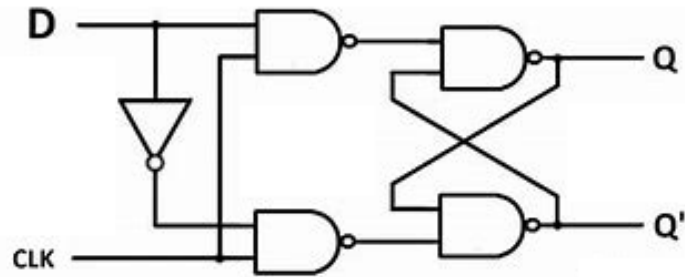
Excitation Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Express S and R in terms of Q_n and Q_{n+1}

D Flip Flop



D	S	R	Q	State
	0	0	Previous State	No Change
0	0	1	0	Reset
1	1	0	1	Set
	1	1	?	Forbidden

SR & D Flip Flop TruthTable

CLK	D	Q	Q(not)
↑	0	0	1
↑	1	1	0

Characteristic Table of D Flip Flop

The characteristic table displays the behavior of the flip flop for each combination of input and current state. The characteristic table for a D flip flop is as follows.

CLK	D	Q	Q(not)
↑	0	0	1
↑	1	1	0

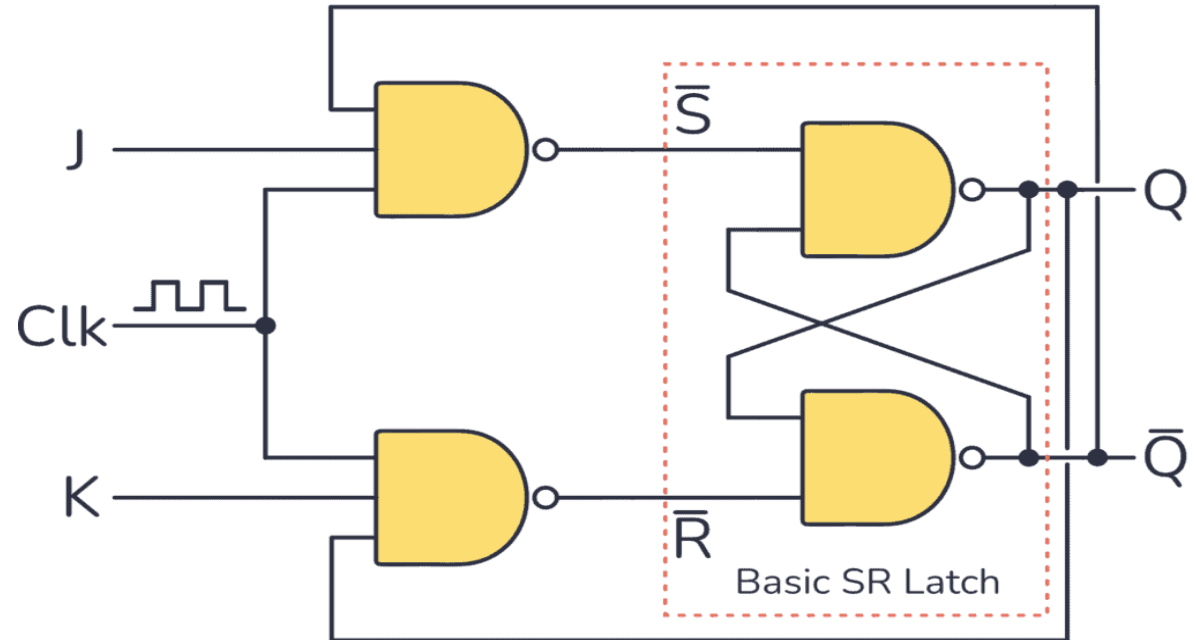
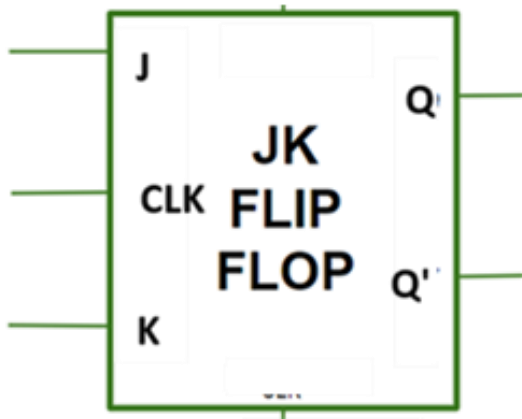
D	Q(Current)	Q(n+1) (Next)
0	0	0
0	1	0
1	0	1
1	1	1

D Flip Flop Excitation Table

D	Q(Current)	Q(n+1) (Next)
0	0	0
0	1	0
1	0	1
1	1	1

Q(Current)	Q(n+1) (Next)	D
0	0	0
1	0	0
0	1	1
1	1	1

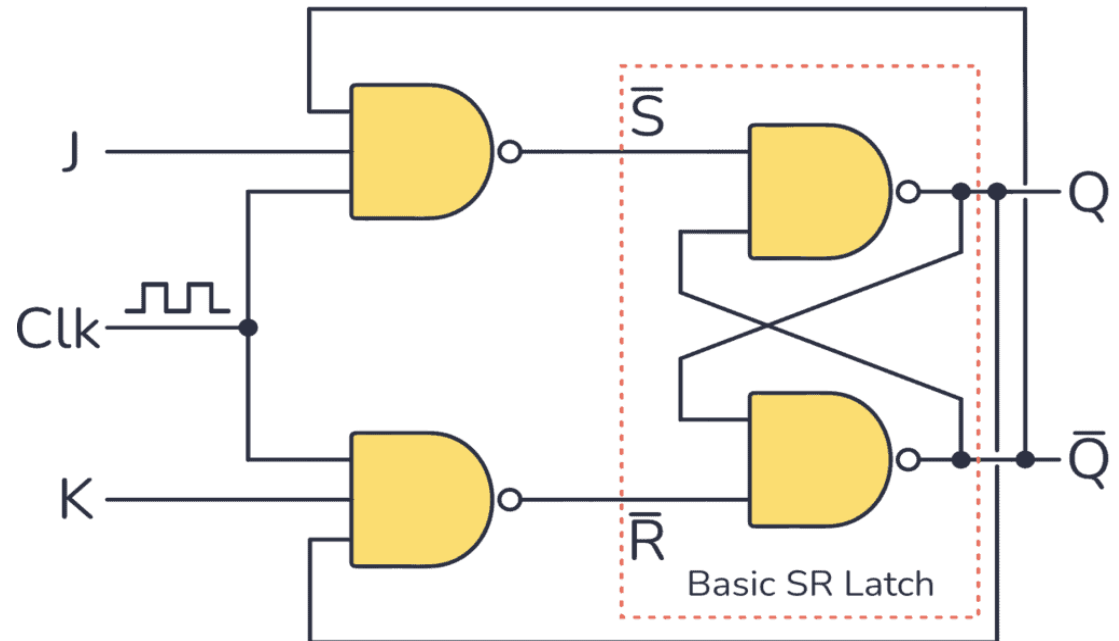
J K Flip Flop



J K Flip Flop

SR Latch Truth table

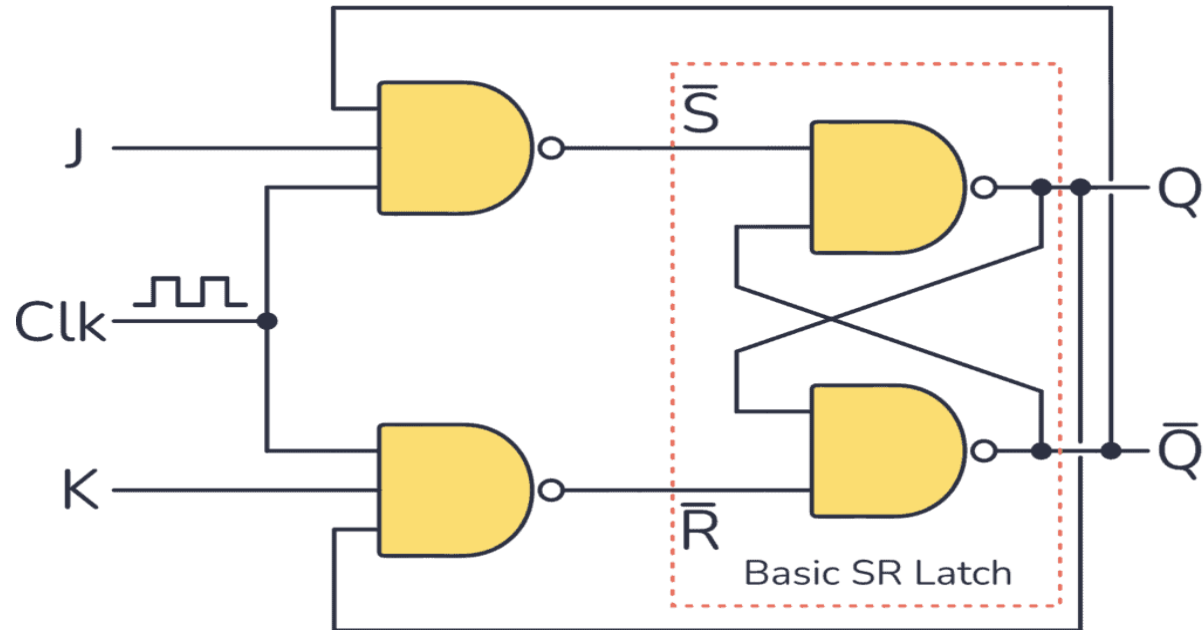
$\overline{S}$	$\overline{R}$	Q	Q'
0	0	Invalid	
0	1	1	0
1	0	0	1
1	1	No change	



CLK	S	R	Q	$\overline{Q}$
0	X	X	Hold	
1	0	0	Hold	
1	0	1	0	1
1	1	0	1	0
1	1	1	Invalid	

Clock	J	K	Q	$\overline{Q}$
0	X	X	Hold	
1	0	0	Hold	
1	0	1	0	1
1	1	0	1	0
1	1	1	Toggle	

J K Flip Flop Truth table



Clock	J	K	Q	$\bar{Q}$
0	X	X	Hold	
1	0	0	Hold	
1	0	1	0	1
1	1	0	1	0
1	1	1	Toggle	

Clock	J	K	Q_{n+1}	State
0	X	X	Q_n	Hold
1	0	0	Q_n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	$\bar{Q}_n$	Toggle

J K Flip Flop – Characteristic Table



INPUT			OUTPUT
J	K	Q_n	$Q(n+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Clock	J	K	Q_{n+1}	State
0	x	x	Q_n	Hold
1	0	0	Q_n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	$\bar{Q}_n$	Toggle

		KQ_n			
J		$\bar{K}\bar{Q}_n$	$\bar{K}Q_n$	KQ_n	$K\bar{Q}_n$
$\bar{J}$		0	1	0	0
J		1	1	0	1

		KQ_n			
J		00	01	11	10
0	0	0	1	0	0
1	1	1	1	0	1

Characteristic Equation: $\bar{K}Q_n + J\bar{Q}_n$

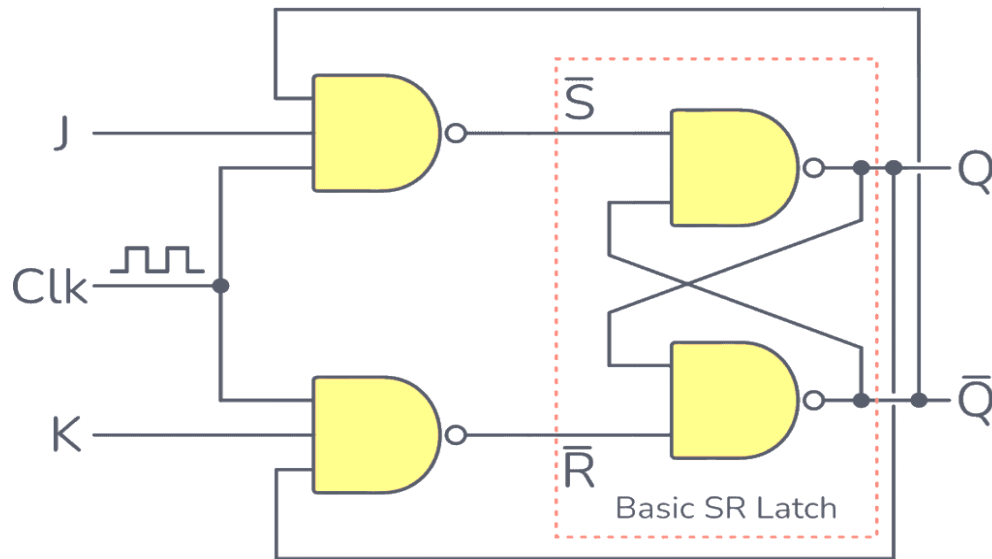
J K Flip Flop - Excitation Table

INPUT			OUTPUT
J	K	Q_n	$Q(n+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

INPUT		OUTPUT	
Q_n	$Q(n+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Find the expression of J and K in terms of Q_n and Q_{n+1}

Race around condition of J K Flip Flop/ realize/dia



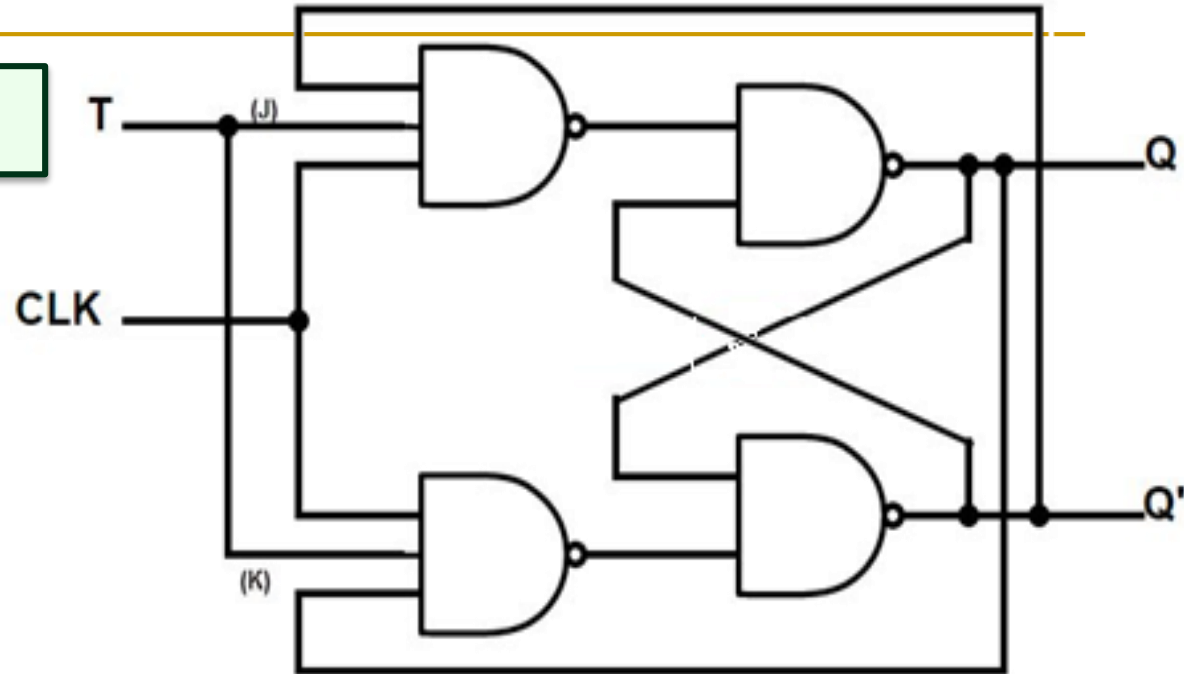
Clock	J	K	Q	$\bar{Q}$
0	X	X	Hold	
1	0	0	Hold	
1	0	1	0	1
1	1	1	1	0
1	1	1	Toggle	

T (Toggle)Flip Flop

Clock	J	K	Q	$\overline{Q}$
0	X	X	Hold	
1	0	0	Hold	
1	0	1	0	1
1	1	0	1	0
1	1	1	Toggle	



CLK	T	Q_{n+1}
↑	0	Q_n
↑	1	Q_n'



Truth Table of T Flip Flop

T	Q_n	Q_{n+1}
0	0	0
1	0	1
0	1	1
1	1	0

T Flip Flop - characteristic Table and Equation

T	Q_n	Q_{n+1}
0	0	0
1	0	1
0	1	1
1	1	0

		Q_n	Q_n'	Q_n
			0	1
T	Q_n			
	T' 0			1
	T 1	1		

$$Q(n+1) = TQ_n' + T'Q_n = T \text{ XOR } Q_n$$

T Flip Flop – Excitation Table

T	Q_n	Q_{n+1}
0	0	0
1	0	1
0	1	1
1	1	0



Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	1	0
1	0	1

$$T = Q_n' Q_{n+1} + Q_n Q_{n+1}'$$